Weiyi He

619 Red Cedar Road, Michigan State University, East Lansing, MI 48824
+1 (517) 512-0022 heweiyi@msu.edu www.linkedin.com/in/weiyi-he github.com/hwyii hwyii.github.io

EDUCATION

Michigan State University **Aug. 2024 – Present**
Ph.D. in Statistics, Dual Major in Computer Science *East Lansing, MI*

University of Science and Technology of China (USTC) **Sep. 2020 – Jul. 2024**
B.S. in Statistics (Outstanding Graduate Award) *Hefei, China*

WORK EXPERIENCE

Amazon.com, Inc. **Jun. 2026 – Aug. 2026**
Applied Scientist Intern, AGI Foundations *Arlington, VA*

- Proposed a novelty-search framework that uses LLM-based mutation and crossover to generate difficulty-controlled and diverse perturbations of mathematical problems, substantially outperforming six existing perturbation methods in diversity and difficulty, especially on strong post-trained models.
- Led a study of input- and weight-space robustness across 25+ open LLMs, finding a strong cross-model correlation. Ran GRPO and SFT interventions on 7 models, showing that both improve input robustness but move weight robustness in opposite directions. First-author manuscript in preparation.

Articuler AI, Inc. **Jun. 2024 – Aug. 2024**
Generative AI Development Intern (Remote) *San Francisco, CA*

- Spearheaded the development of the **Search Agent** product — a scalable Python pipeline that automates professional background research for end users, from data collection to structured insight generation. (Deployed to production)
- Designed multi-step LLM workflows for profile parsing, query generation, information extraction, and validation using Google Search APIs and custom web scraping, improving retrieval quality and user experience.

Elven Technologies Pte. Ltd. **Jan. 2024 – May. 2024**
Data Engineering Intern (Remote) *Singapore*

- Built Python API integrations with major cryptocurrency exchanges (e.g., Coinbase, Binance, OKX) to automate transaction extraction and synchronization for Elven's Web3 financial reporting platform.
- Parsed and formatted transaction logs, enabling reliable downstream reporting and visualization in the platform.

RESEARCH AND OTHER PROJECTS

ReFT-based latent adversarial training on low-dimensional manifolds **Dec. 2025 – Apr. 2026**

- Developed a defense strategy combining Low-rank Representation Fine-tuning (LoReFT) with Latent Adversarial Training (LAT); significantly mitigated Greedy Coordinate Gradient (GCG) attacks by implementing a multi-position intervention mechanism that prevents adversarial signal accumulation across causal sequences.
- Engineered a **Circuit-Aware** training framework utilizing Attribution Patching to isolate critical model circuits; optimized the inner-loop optimization by dynamically pruning non-essential gradient paths, achieving a 30% reduction in backward pass FLOPs while maintaining robust defensive performance.

Privacy Risks in LLM Agent Memory **Oct. 2024 – Feb. 2025**

- Studied privacy risks in LLM agent memory and proposed **MEXTRA (Memory EXTRaction Attack)**, a black-box framework using targeted attacks and automated prompt generation under varying attacker knowledge.
- Evaluated memory vulnerabilities across real-world agent architectures and deployment settings; MEXTRA extracted at least one private item in **83%–90%** of targeted trials across two agents.

PUBLICATIONS & MANUSCRIPTS

- **Weiyi He**, Yuping Lin, Jiliang Tang, Yue Xing. "Efficient LLM Adversarial Training via Low-Rank Defense and Circuit-Guided Surrogates." *arXiv preprint arXiv:2607.28959*, 2026.
- **Weiyi He**, Yue Xing. "Impact of Positional Encoding: Clean and Adversarial Rademacher Complexity for Transformers Under In-Context Regression." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- Bo Wang, **Weiyi He**, Shenglai Zeng, Zhen Xiang, Yue Xing, Jiliang Tang, and Pengfei He. "Unveiling Privacy Risks in LLM Agent Memory." *Proceedings of the Association for Computational Linguistics (ACL)*, 2025.

TECHNICAL SKILLS

Languages & Tools: Python, R, C, Bash, Git, Slurm, Linux
Frameworks: Pytorch, Hugging Face Transformers, verl, vLLM, AWS Bedrock, DeepSeed
ML/Systems: Distributed Training, Supervised Fine-tuning, Reinforcement Learning